Prill-Exceptional Étale Covers from Raynaud Bundles

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Abstract

Let C be a smooth projective connected complex curve of genus $g \geq 2$. We prove that there is a connected finite étale Galois cover $f\colon X \to C$ such that

$$h^0\big(X, \mathcal{O}_X\big(f^{-1}(p)\big)\big) \geq 2$$

for every $p \in C$. The construction starts from the Raynaud bundle R_g, of rank g^g and slope 1. After a determinant-normalising twist, this bundle has no global sections, retains Raynaud's non-vanishing after twisting by any point, and is trivialised by a finite étale cover. An irreducible finite-monodromy summand then has the same pointwise non-vanishing property. The regular representation of its monodromy group produces the required cover.

1 Introduction

Let $f\colon X \to C$ be a finite morphism of smooth projective connected complex curves. Following Prill, one asks whether a fibre divisor $f^{-1}(p)$ can move in a pencil, equivalently whether

$$h^0\big(X, \mathcal{O}_X\big(f^{-1}(p)\big)\big) > 1$$

for a general point $p \in C$. The question appears in [1, Chapter VI, Exercise D] and is recorded, for bases of genus at least 3, as [4, Problem 14]. Landesman and Litt constructed a connected finite étale cover of degree 36 with this property over every curve of genus 2 [3].

A cover with this property at every point will be called *Prill exceptional*. We give a uniform construction in every genus at least 2.

Theorem 1.1. *Let C be a smooth projective connected complex curve of genus $g \geq 2$. There exists a connected finite étale Galois cover $f\colon X \to C$ such that*

$$h^0\big(X, \mathcal{O}_X\big(f^{-1}(p)\big)\big) \geq 2$$

for every $p \in C$.

For $g \geq 3$, Theorem 1.1 gives an affirmative answer to [4, Problem 14].

For a finite flat morphism, the function $p \mapsto h^0(X, \mathcal{O}_X(f^{-1}(p)))$ is upper semicontinuous. Thus a lower bound holding on a dense open subset automatically holds at every point. The pointwise form of Theorem 1.1 is therefore equivalent to the usual "general fibre" formulation, but it is the form naturally delivered by the construction.

The proof uses two features of Raynaud's bundles. First, R_g has a section after tensoring by any degree-zero line bundle. Second, the pullback of R_g under the multiplication-by-g isogeny of the Jacobian is a direct sum of copies of one line bundle. A suitable twist makes the determinant trivial. The latter line bundle then becomes torsion after restriction to the relevant étale pullback of C, so the twisted Raynaud bundle has finite monodromy.

2 The Raynaud bundle

Write $J = \mathrm{Jac}(C) = \mathrm{Pic}^0(C)$, fix a theta divisor Θ on J, and choose an Abel–Jacobi embedding $i\colon C \hookrightarrow J$. We use the following standard form of Raynaud's construction; see [5, Section 3] and [2, Section 2.3].

Theorem 2.1 (Raynaud)**.** *For each integer $n \geq 1$, there is a vector bundle E_n on J such that*

$$[n]^* E_n \cong H^0(J, \mathcal{O}_J(n\Theta))^\vee \otimes_{\mathbf{C}} \mathcal{O}_J(n\Theta). \tag{2.1}$$

Its restriction $R_n = i^* E_n$ is semistable, has rank n^g and slope g/n, and satisfies

$$H^0(C, R_n \otimes \alpha) \neq 0 \qquad \textit{for every } \alpha \in \mathrm{Pic}^0(C). \tag{2.2}$$

We take $n = g$ and set

$$R := R_g, \qquad r := \mathrm{rk}(R) = g^g.$$

Then R is semistable of slope 1, hence $\deg(R) = r$.

3 A determinant-normalised twist

The next lemma isolates the only counting argument in the proof.

Lemma 3.1. *There is a line bundle $M \in \mathrm{Pic}^{-1}(C)$ such that*

$$M^{\otimes r} \cong (\det R)^{-1} \qquad \textit{and} \qquad H^0(C, R \otimes M) = 0. \tag{3.1}$$

Proof. The map

$$\mathrm{Pic}^{-1}(C) \longrightarrow \mathrm{Pic}^{-r}(C), \qquad M \longmapsto M^{\otimes r},$$

is a translate of the multiplication-by-r isogeny of J. Consequently the fibre

$$T := \left\{ M \in \mathrm{Pic}^{-1}(C) : M^{\otimes r} \cong (\det R)^{-1} \right\}$$

has cardinality r^{2g}.

Call $M \in \mathrm{Pic}^{-1}(C)$ bad if $H^0(C, R \otimes M) \neq 0$. A nonzero section determines a nonzero morphism

$$M^{-1} \longrightarrow R.$$

Let $L \subset R$ be the saturation of its image. Then $L \cong M^{-1}(D)$ for some effective divisor D. Since $\deg(M^{-1}) = 1$ and R is semistable of slope 1, one has

$$1 + \deg D = \deg L \leq 1.$$

Thus $D = 0$, and M^{-1} is a degree-1 line subbundle of R.

Fix a Jordan–Hölder filtration of R. Any degree-1 line subbundle $L \subset R$ is stable and has the same slope as R. Choose the first term of the filtration containing L. The induced nonzero map from L to the corresponding stable quotient is an isomorphism, because non-isomorphic stable bundles of the same slope admit no nonzero morphisms. Hence L is isomorphic to a rank-one Jordan–Hölder factor of R. There are at most r such factors, counted with multiplicity. It follows that there are at most r isomorphism classes of bad M.

Since $g \geq 2$ and $r = g^g > 1$, we have $r^{2g} > r$. Therefore the set T contains an element that is not bad. $\square$

Fix M as in Lemma 3.1 and put

$$V := R \otimes M.$$

Then

$$\deg V = 0, \qquad \det V \cong \mathcal{O}_C, \qquad H^0(C, V) = 0. \tag{3.2}$$

For every $p \in C$, the line bundle $M(p)$ has degree zero, so (2.2) gives

$$H^0(C, V(p)) = H^0(C, R \otimes M(p)) \neq 0. \tag{3.3}$$

4 Finite monodromy

Proposition 4.1. *The vector bundle V is trivialised by a connected finite étale cover of C.*

Proof. Form the cartesian square

$$\begin{array}{ccc} \widetilde{C} & \xrightarrow{\ \widetilde{i}\ } & J \\ {\scriptstyle q}\big\downarrow & & \big\downarrow{\scriptstyle [g]} \\ C & \xrightarrow{\ i\ } & J. \end{array}$$

The morphism q is finite étale and surjective. By (2.1),

$$\begin{aligned} q^*V &\cong \widetilde{i}^*[g]^*E_g \otimes q^*M \\ &\cong H^0(J, \mathcal{O}_J(g\Theta))^\vee \otimes_{\mathbf{C}} \left(\widetilde{i}^*\mathcal{O}_J(g\Theta) \otimes q^*M\right). \end{aligned}$$

Let $\widetilde{C}_0$ be a connected component of $\widetilde{C}$. Its image under q is nonempty, open, and closed; hence $q|_{\widetilde{C}_0}$ is surjective because C is connected. Write

$$N := \left(\widetilde{i}^*\mathcal{O}_J(g\Theta) \otimes q^*M\right)\Big|_{\widetilde{C}_0}.$$

Since $\dim H^0(J, \mathcal{O}_J(g\Theta)) = g^g = r$, the restriction of q^*V to $\widetilde{C}_0$ is $N^{\oplus r}$. Taking determinants and using $\det V \cong \mathcal{O}_C$, we obtain

$$N^{\otimes r} \cong \mathcal{O}_{\widetilde{C}_0}.$$

Thus N is torsion. Let d be its order and choose an isomorphism $N^{\otimes d} \cong \mathcal{O}_{\widetilde{C}_0}$. The relative spectrum

$$\mathrm{Spec}_{\widetilde{C}_0}\left(\bigoplus_{a=0}^{d-1} N^{-a}\right)$$

with multiplication induced by this isomorphism is a finite étale μ_d-torsor, and the pullback of N to it is trivial. Any connected component maps surjectively to $\widetilde{C}_0$. Composing such a component with $\widetilde{C}_0 \to C$ gives a connected finite étale morphism $h \colon C' \to C$ for which h^*V is trivial. $\square$

Taking the Galois closure of this finite étale cover, if necessary, gives another finite étale cover and shows that V is the vector bundle associated with a finite-dimensional complex representation of a finite group.

Proposition 4.2. *There is a nontrivial irreducible finite-monodromy vector bundle W on C such that*

$$H^0(C, W(p)) \neq 0 \qquad \textit{for every } p \in C.$$

Proof. Choose a connected finite étale Galois cover $u\colon Y \to C$ trivialising V, with Galois group H. Fix a trivialisation $u^*V \cong \mathcal{O}_Y^{\oplus r}$. For each $\gamma \in H$, the comparison between this trivialisation and its pullback by γ is an automorphism of $\mathcal{O}_Y^{\oplus r}$. Its matrix entries lie in $H^0(Y, \mathcal{O}_Y) = \mathbf{C}$, since Y is connected and projective. The descent datum is therefore a representation of H, and V is its associated vector bundle. By Maschke's theorem,

$$V \cong \bigoplus_{j=1}^{s} W_j^{\oplus m_j},$$

where the W_j are pairwise non-isomorphic irreducible finite-monodromy bundles. No W_j is trivial: a trivial summand would give a nonzero global section of V, contrary to (3.2).

For each j, define

$$Z_j := \left\{ p \in C : H^0(C, W_j(p)) \neq 0 \right\}.$$

Let $p_1, p_2\colon C \times C \to C$ be the projections and let $\Delta \subset C \times C$ be the diagonal. The restriction of

$$p_1^* W_j \otimes \mathcal{O}_{C\times C}(\Delta)$$

to $C \times \{p\}$ is $W_j(p)$. Properness of p_2 and upper semicontinuity therefore show that Z_j is closed.

Equation (3.3) implies $C = \bigcup_{j=1}^{s} Z_j$. Since C is irreducible and the union is finite, one of the Z_j equals C. Taking W to be the corresponding W_j proves the proposition. □

5 The covering

Proof of Theorem 1.1. Let W be as in Proposition 4.2, and let

$$\rho\colon \pi_1(C) \longrightarrow \mathrm{GL}(W_c)$$

be its monodromy representation. Its image G is finite. The subgroup $\ker \rho$ determines a connected finite étale Galois cover

$$f\colon X \longrightarrow C$$

with Galois group G.

The vector bundle $f_*\mathcal{O}_X$ is associated with the regular representation $\mathbf{C}[G]$. Every irreducible complex representation of G occurs in $\mathbf{C}[G]$, with multiplicity equal to its dimension. In particular, $\mathcal{O}_C$ and W occur as distinct direct summands of $f_*\mathcal{O}_X$.

For $p \in C$, the divisor $f^{-1}(p)$ is the pullback of p, so

$$\mathcal{O}_X\big(f^{-1}(p)\big) \cong f^*\mathcal{O}_C(p).$$

By the projection formula,

$$\begin{aligned} H^0\big(X, \mathcal{O}_X\big(f^{-1}(p)\big)\big) &\cong H^0(C, f_*\mathcal{O}_X \otimes \mathcal{O}_C(p)) \\ &\supset H^0(C, \mathcal{O}_C(p)) \oplus H^0(C, W(p)). \end{aligned}$$

The first summand is nonzero because p is effective, and the second is nonzero by Proposition 4.2. Hence the left-hand side has dimension at least 2 for every $p \in C$. □

Remark 5.1. The construction is not degree-effective. The first bundle already has rank g^g, and the proof subsequently passes to an étale Kummer cover and a Galois closure. The degree-36 construction of [3] in genus 2 is therefore much sharper numerically.

Remark 5.2. The determinant-normalising step is essential. Formula (2.1) only makes the pullback of R_g a direct sum of copies of a line bundle. Trivial determinant forces that line bundle to be torsion, which is what converts projective finite monodromy into finite linear monodromy.

References

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